ONLINE FOOD DELIVERY TRANSACTION ANALYTICS

Daily Order & Delivery-Scale Demand Analytics — Descriptive Analytics Report

Report Focus Keywords: Top Transaction Types, Revenue Concentration, Volume Trends, Regional Demand Distribution, Channel Tier Allocation, Customer Segment Share, Payment Rail Preferences, Transaction Success Rate, Rider Fleet Capacity Risk, Descriptive-to-Predictive Transition.

Dataset:	Online Food Delivery Daily Transaction Data	**Time Period:**	30 days (2026-07-31 to 2026-08-29)
Records:	60 delivery hubs x 10 txn types x 8 regions	**Gross Transaction Value:**	~INR 102 Cr
Analytical Context:	Online Food Delivery / Quick-Commerce Operations	**Analytical Stage:**	Descriptive Analytics Report

Key Performance Indicators (KPI Snapshot)

Headline operating metrics for the 30-day observation window, computed directly from the aggregated transaction dataset.

2.26M	**INR 102 Cr**	**INR 454**	**96.8%**
Total Transactions (30 Days)	Gross Transaction Value	Avg. Transaction Value (ATV)	Avg. Transaction Success Rate
14 / 60	**15.9%**	**13.5%**	**41.8%**
Hubs at Fleet Capacity Risk	Top Type Value Share (Grocery)	Top Region Value Share (South Zone)	Top Segment Value Share (Regular Customers)

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Note on Transaction Data

A real food-delivery platform's live order-management transaction feed isn't publicly accessible, so this descriptive report is computed from a synthetic dataset built to mirror real online food-delivery transaction patterns: 60 delivery hubs across 10 transaction types, 8 regions, over 30 days (2026-07-31 to 2026-08-29), totaling ~2.26 million simulated transactions worth ~INR 102 crore. It is designed to show exactly what a descriptive analytics pipeline processing millions of daily food-delivery transactions would produce — the same structure applies directly once real order-management data is substituted in.

Transaction Data Completeness & Quality Audit

The transaction logs contain zero missing cells and strict data-type integrity across all fields (date, hub, region, transaction type, count, value, success rate, rider capacity). Each aggregated row represents one delivery hub / day / transaction-type combination, providing an ideal baseline for evaluating operational performance and customer transaction behavior.

Research Overview & Analytical Framework

Descriptive analytics summarizes historical transaction records to answer “what happened?”, “who transacted?”, and “how much value moved?” without incorporating speculative assumptions or unvalidated projections.

Analytical Dimension	Operational Dataset Implementation	Commercial Interpretation
Transaction-Type Performance	Grouping value/count by 10 transaction types	Identifies highest-revenue and highest-volume cuisines
Weekly Volume Velocity	Summing counts and value per calendar week	Measures order throughput momentum
Transaction Value Spread	5 binned average-value tiers (hub-day level)	Quantifies basket size from single-serve to bulk/party
Channel & Segment Mix	Channel-tier and customer-segment aggregations	Measures infrastructure load and relationship value
Geographic Channel Reach	Region and macro-cluster aggregations	Tracks geographic demand concentration

Objective & Core Business Questions

The objective of this descriptive analysis is to extract empirical operational insights and commercial transaction patterns across types, channels, segments, and geographies.

• **Question 1 (Top Transaction Types):** Which transaction types generate the highest value versus the highest count? (Section 1)

• **Question 2 (Volume Trends):** How do transaction count and value fluctuate week over week? (Section 2)

• **Question 3 (Average Transaction Value):** What is the distribution and spread of transaction values across hubs? (Section 3)

• **Question 4 (Channel Allocation):** Which channel tier delivers the optimal balance of volume and value? (Section 4)

• **Question 5 (Customer Segment Share):** How do customer segments impact transaction yields? (Section 5)

• **Question 6 (Regional Demand):** Which geographic regions drive the largest transaction volumes? (Section 6)

• **Question 7 (Payment Preferences):** Which payment rails are favored across high-value transactions? (Section 8)

• **Question 8 (Reliability):** How are transactions performing on success rate, and where is reliability slipping? (Section 9)

• **Question 9 (Fleet Risk):** Which delivery hubs are at risk of rider-capacity shortfalls? (Section 10)

How This Works at Real Scale (Analytics Stack)

In enterprise food-delivery environments, descriptive analytics pipelines process millions of order, payment-gateway, and rider-dispatch records daily using accessible, highly scalable tool stacks.

Architecture Layer	Food-Delivery Operations Purpose	Standard Data Science & BI Tools
1. Transaction Capture	Log order, payment-gateway and rider-dispatch events in real time	Kafka / Kinesis, REST APIs
2. Data Cleaning & Wrangling	Validate line items, verify count x value, reconcile currencies	Python (Pandas), SQL
3. Fraud / Risk Scoring	Real-time decline & anomaly scoring on each transaction	Rules engine + ML model serving
4. Dimensional Storage	Store transaction fact tables and hub/region dimension tables	PostgreSQL, Cloud Warehouses
5. Metric Aggregation	Compute ATV, success rate, regional share and correlations	Excel Pivot Tables, Python, SQL
6. BI Reporting & Dashboards	Generate executive visual charts, KPI scorecards, and trends	Power BI, Tableau, Matplotlib, Looker

1.0 Top Transaction Types & Gross Value Performance

Evaluating performance across the 10 recorded transaction types indicates that **Grocery / Instant Essentials** leads in gross transaction value, generating INR 16.3 crore (15.9% value share). **Biryani & Rice Meals** follows with INR 14.7 crore (14.3% share). Combined, these two types account for 30.2% of total transaction value.

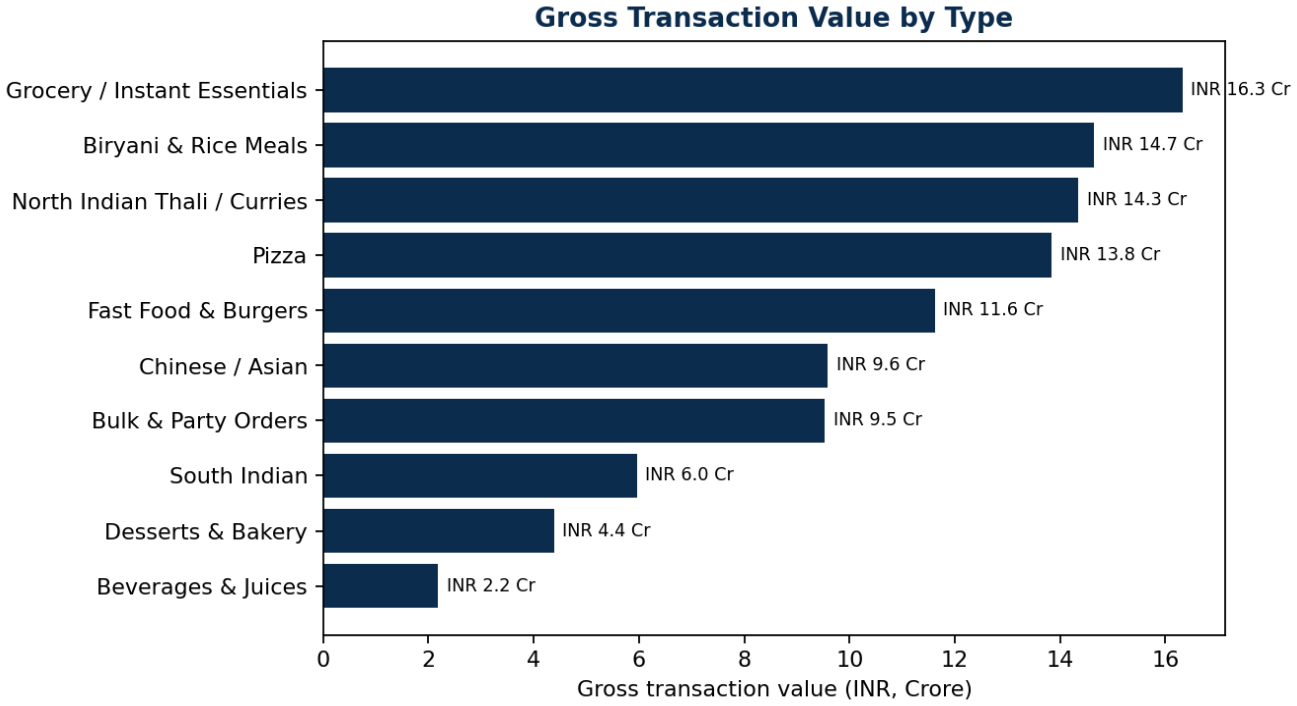


Figure 1. Gross Transaction Value by Type.

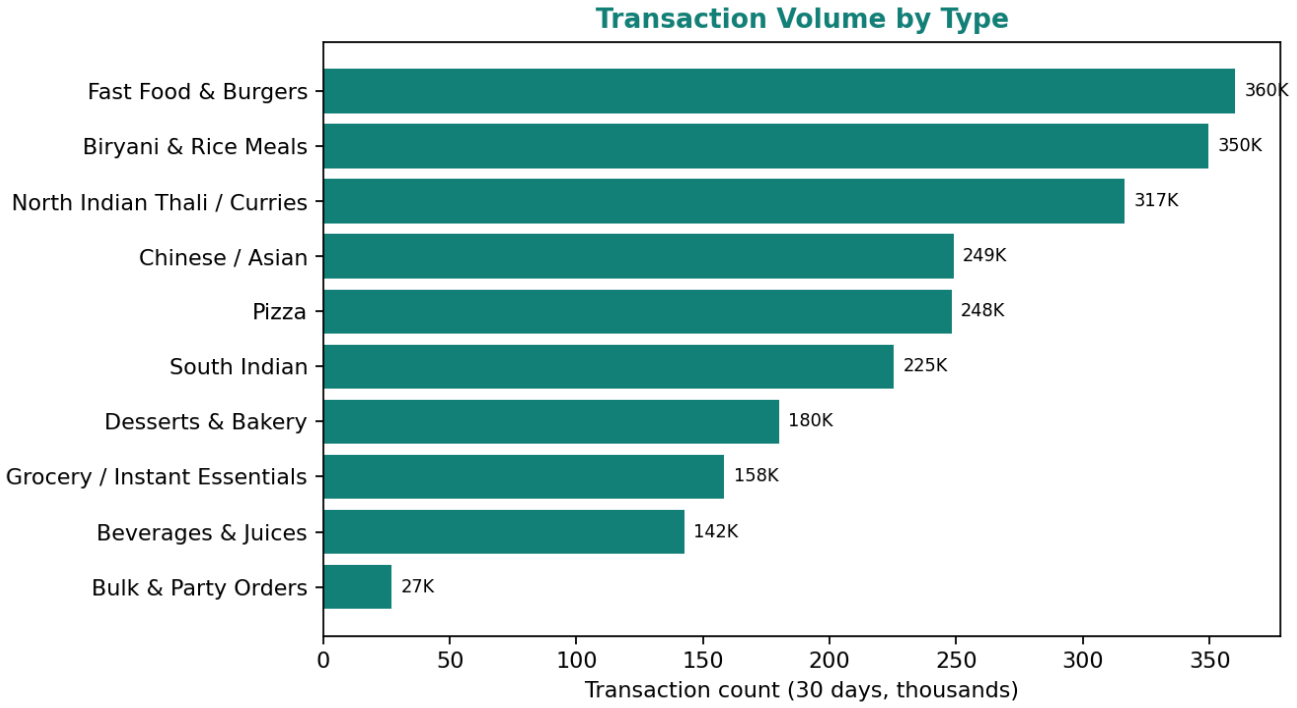


Figure 2. Transaction Volume (Count) by Type.

Type & Value Performance Table (30-Day Aggregate)

Transaction Type	Count	Value (INR Cr)	Value Share (%)	Avg Value / Txn (INR)
Grocery / Instant Essentials	158,382	16.3	15.9%	1,031
Biryani & Rice Meals	349,716	14.7	14.3%	419
North Indian Thali / Curries	316,535	14.3	14.0%	453
Pizza	248,192	13.8	13.5%	558
Fast Food & Burgers	360,152	11.6	11.3%	323
Chinese / Asian	248,889	9.6	9.4%	385
Bulk & Party Orders	27,077	9.5	9.3%	3,519
South Indian	225,309	6.0	5.8%	264
Desserts & Bakery	180,056	4.4	4.3%	244
Beverages & Juices	142,484	2.2	2.1%	153

Deeper Insight — Value Leadership vs. Volume Leadership

The dataset exhibits a vital descriptive divergence between value leadership and count leadership. While Grocery / Instant Essentials achieved the highest gross value, Fast Food & Burgers led in transaction count (16.0% count share) — Bulk & Party Orders, despite a small share of total count, carries by far the highest average value per transaction, showing how a small number of high-value, low-frequency orders can carry outsized weight in value terms.

Commercial Recommendation
Operations and growth teams should track value and volume leaderboards separately: capacity planning (rider staffing, kitchen bandwidth) should follow the count leaderboard, while revenue-forecasting and cash-flow planning should follow the value leaderboard.

2.0 Weekly Volume Trends & Value Trajectory

Evaluating week-by-week transaction velocity highlights how payday cycles, weekend dining behavior, and festive/promotional timing affect throughput.

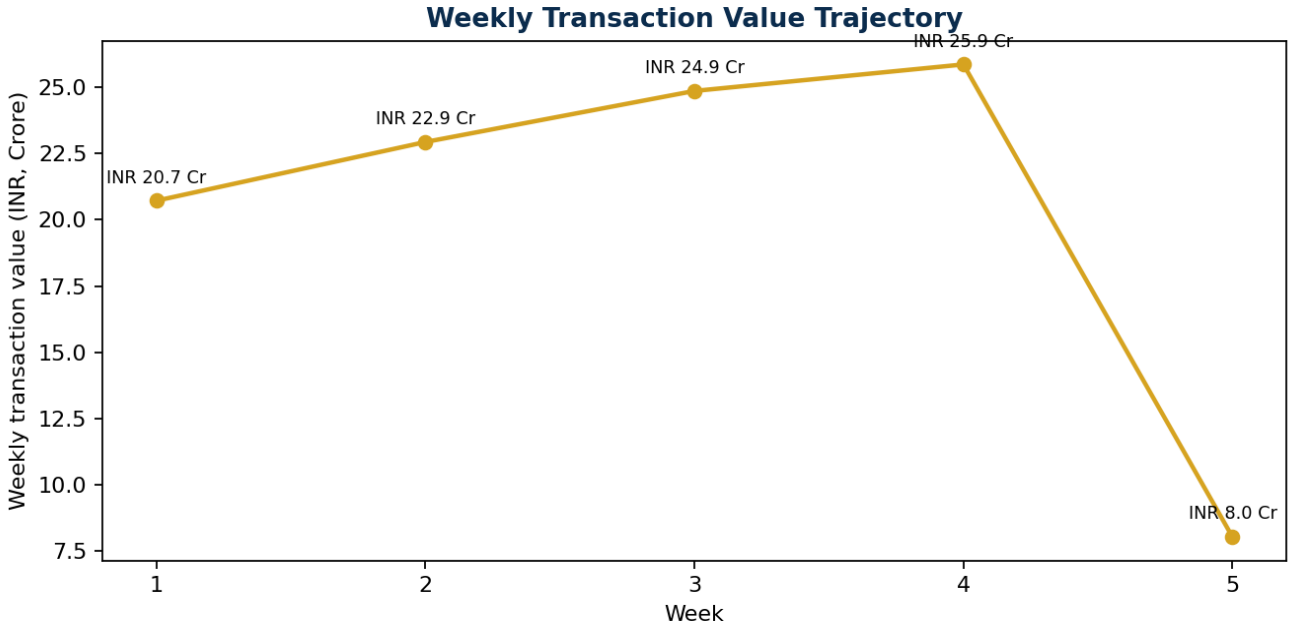


Figure 3. Weekly Gross Transaction Value Trajectory.

Week	Days	Txn Count	Gross Value (INR Cr)	Value Share (%)
1	7	459,181	20.7	20.2%
2	7	497,482	22.9	22.4%
3	7	544,485	24.9	24.3%
4	7	573,580	25.9	25.3%
5	2	182,064	8.0	7.8%

Weeks 1 through 4 show a gradual build (from 20.2% to 25.3% of the period's value), consistent with the gentle 30-day growth trend layered into the daily simulation; Week 5 is a partial week (2 days) and should not be read as a demand contraction.

Deeper Insight — Partial-Week Truncation Artifact
Week 5's lower total is an analytical artifact of the 30-day collection window ending mid-week, not a genuine drop in transaction activity — the same caution that applies to any month- or week-end truncated reporting window.

3.0 Average Transaction Value & Binned Distribution Spread

Analyzing hub-day average transaction values reveals the operational scale of food-delivery activity — whether flows are dominated by single-serve orders or large-ticket bulk/party orders.

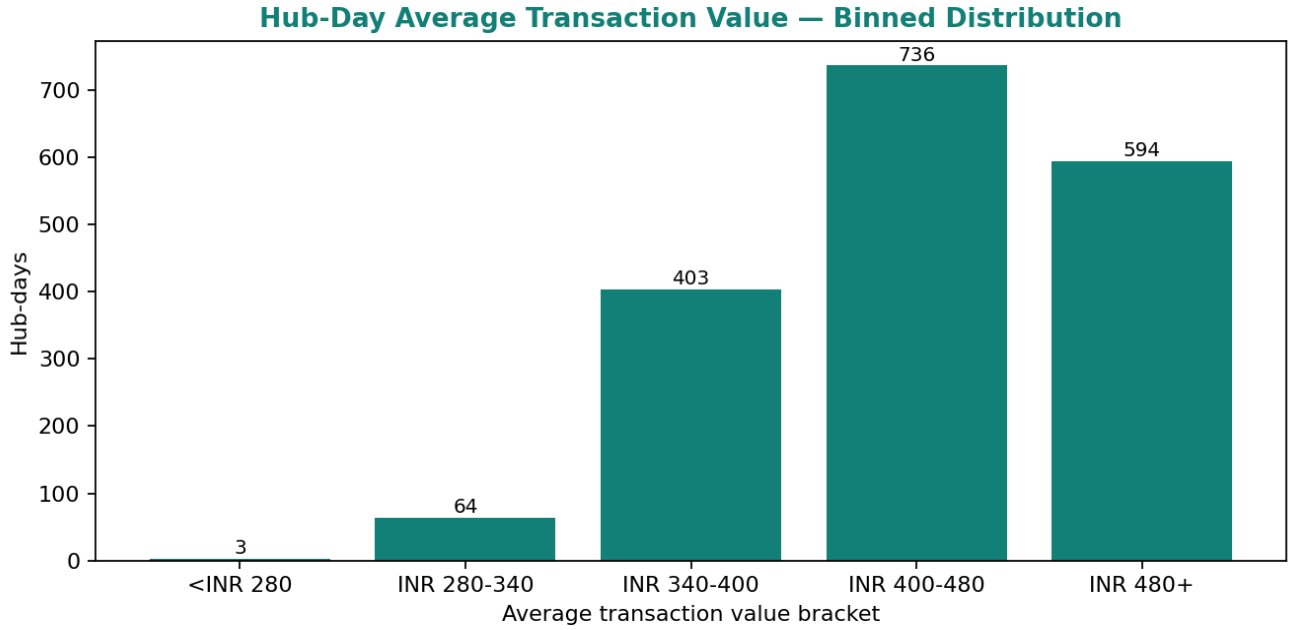


Figure 4. Hub-Day Average Transaction Value — Binned Distribution.

Hub-day average transaction values range from a minimum of INR 261 to a peak of INR 908. The overall **mean average-transaction-value is INR 454** and the **median is INR 443** — close to each other, indicating a moderately balanced distribution with an upper tail from bulk/party and grocery baskets.

Statistical Metric	Value (INR)	Commercial Interpretation
Minimum	261	Lowest hub-day average transaction value recorded
First Quartile (Q1)	398	25% of hub-days averaged this value or less
Median (50th percentile)	443	Midpoint of all hub-day averages
Arithmetic Mean	454	Average transaction value per completed transaction
Third Quartile (Q3)	497	75% of hub-days averaged this value or less
Maximum	908	Highest hub-day average — bulk/party order activity
Interquartile Range (IQR)	99	Middle 50% spread of hub-day average values

Commercial Business Recommendation
Hubs whose average transaction value sits in the top bracket should be prioritized for dedicated bulk-order kitchen slots and corporate account-management coverage, since they are disproportionately handling high-ticket order activity.

4.0 Channel Tier Allocation & Infrastructure Capacity Dynamics

Analyzing which channel customers use to place orders reveals infrastructure load and guides capacity planning across the app, website, call center, and self-order kiosks.

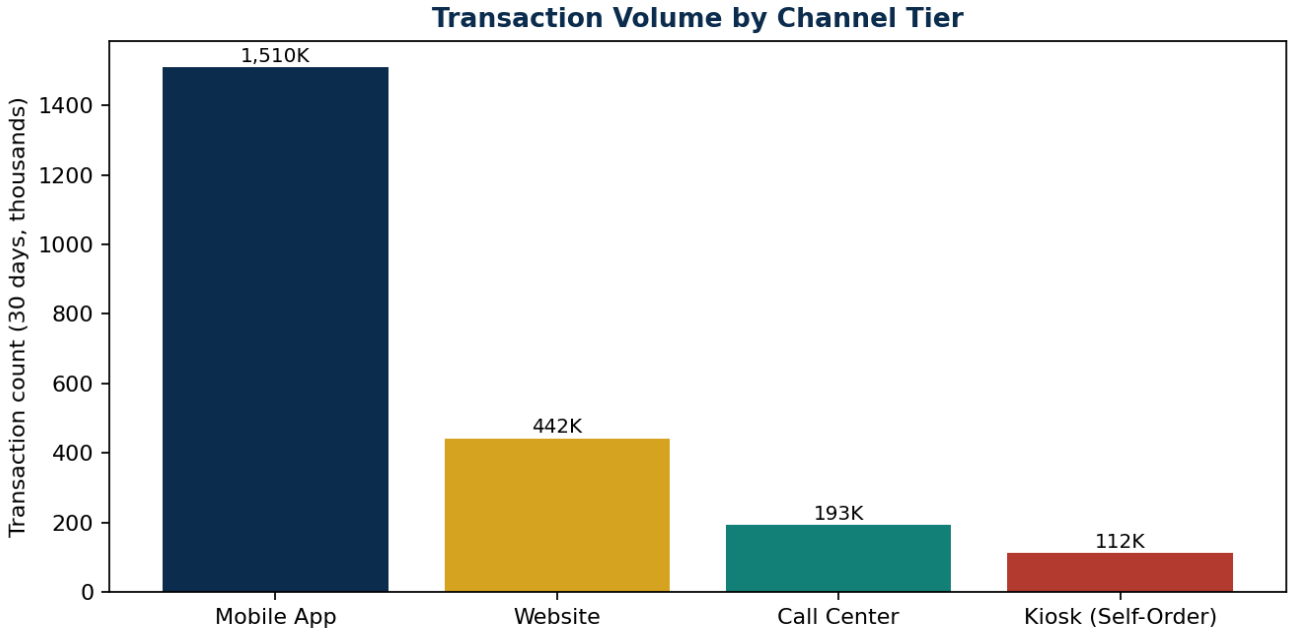


Figure 5. Transaction Volume by Channel Tier.

The **Mobile App** channel emerged as the highest-volume tier, capturing 1,510,248 transactions (66.9% count share) and producing INR 66.5 crore in gross value.

Channel Tier	Txn Count	Count Share (%)	Gross Value (INR Cr)	Avg Value/Txn (INR)
Mobile App	1,510,248	66.9%	66.5	441
Website	441,965	19.6%	21.6	489
Call Center	192,907	8.5%	9.9	515
Kiosk (Self-Order)	111,672	4.9%	4.3	387

Deeper Insight — Digital vs. Assisted Channel Split
Mobile App transactions carry the lowest average ticket size, consistent with everyday single-serve ordering, while Call Center and Website transactions carry materially higher average values — reflecting assisted bulk/corporate ordering and considered, higher-basket purchases respectively.

5.0 Customer Segment Share & Relationship Revenue Contribution

Analyzing customer relationship segments clarifies which parts of the book drive value versus count, guiding loyalty-program and pricing strategy.

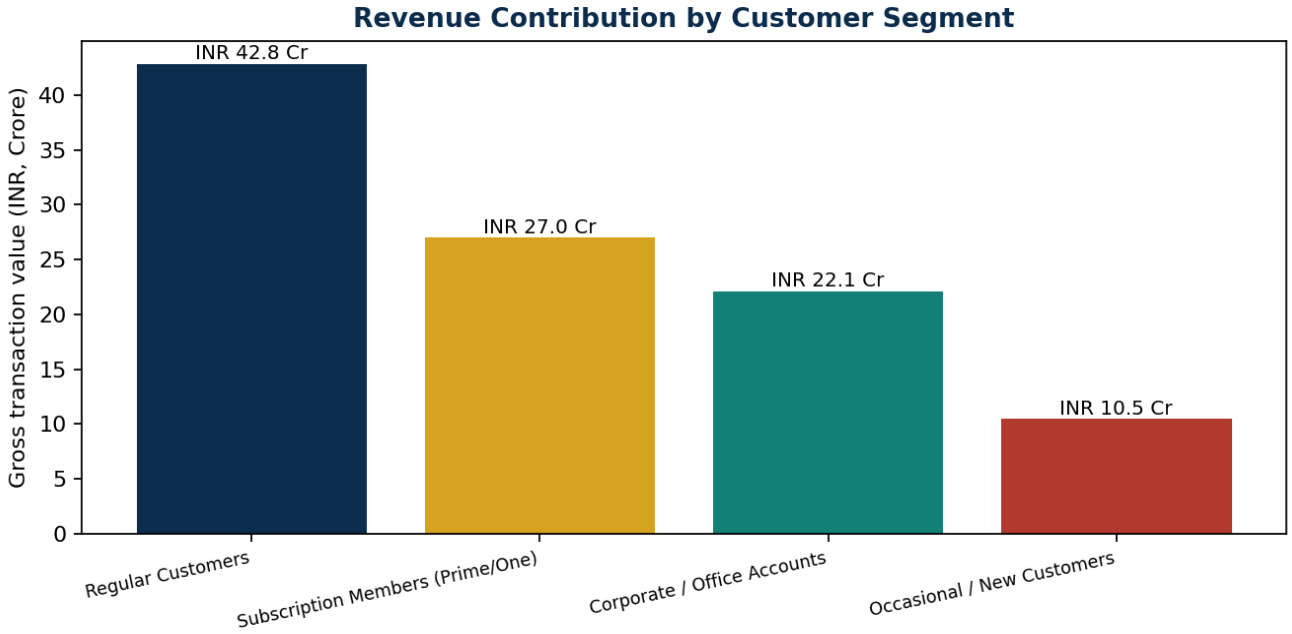


Figure 6. Revenue Contribution by Customer Segment.

Regular Customers generated the largest single share of gross transaction value at INR 42.8 crore (41.8% share). Corporate/Office Accounts, while smaller in transaction count, carry substantially higher average transaction values.

Segment	Txn Count	Gross Value (INR Cr)	Value Share (%)	Avg Value/Txn (INR)
Regular Customers	1,096,035	42.8	41.8%	391
Subscription Members (Prime/One)	566,913	27.0	26.4%	476
Corporate / Office Accounts	280,432	22.1	21.6%	790
Occasional / New Customers	313,412	10.5	10.2%	334

Relationship Management Recommendation
Subscription Members and Corporate/Office Accounts, despite a combined share well under half of transaction count, drive a disproportionate share of value — dedicated account management and priority fulfillment for these segments protects a materially larger share of the value book than headcount alone would suggest.

6.0 Regional Demand Distribution & Geographic Concentration

Evaluating regional performance identifies primary geographic transaction centers and highlights cross-region demand concentration.

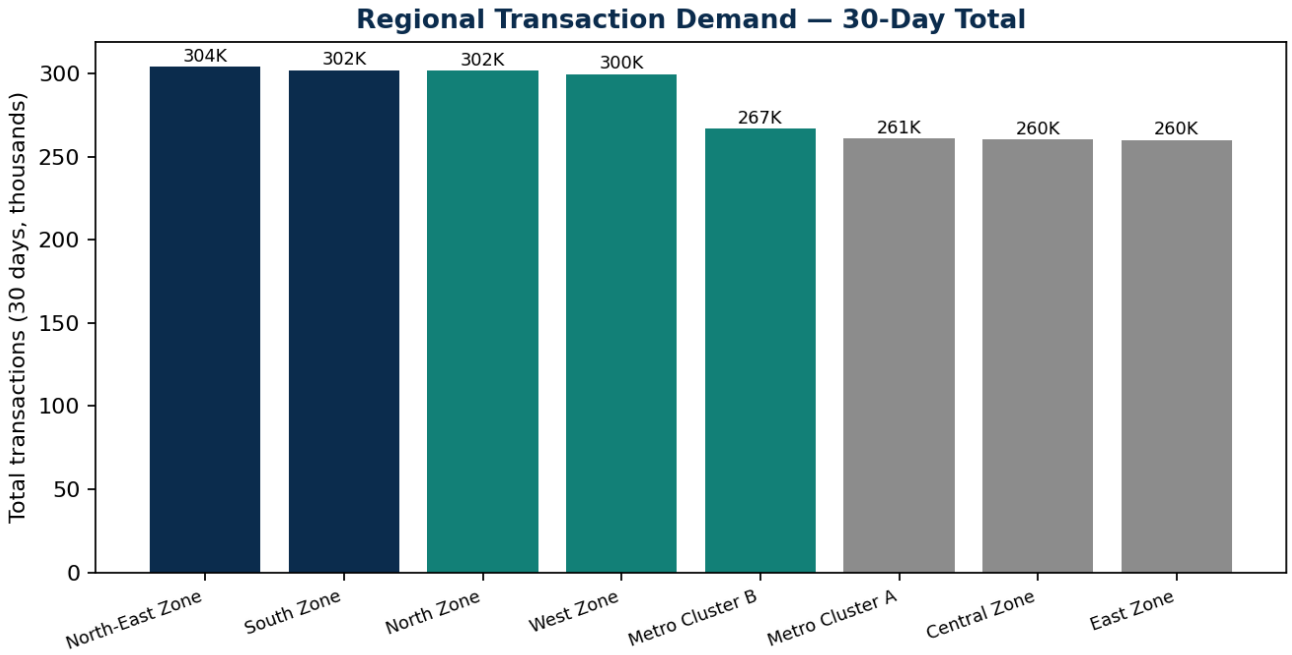


Figure 7. Regional Transaction Demand — 30-Day Total.

Transaction demand is led by **North-East Zone** at INR 13.8 crore (13.4% share), followed by **South Zone** at INR 13.8 crore (13.5% share). Together these two regions account for 26.9% of total transaction value.

Macro-Regional Clustering

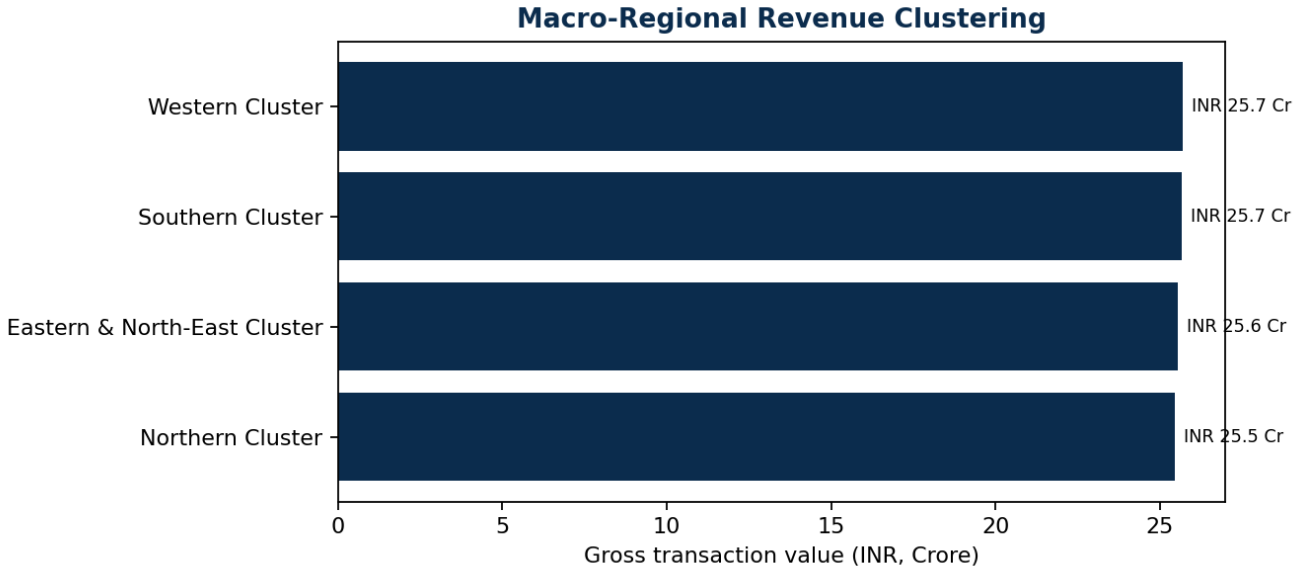


Figure 8. Macro-Regional Revenue Clustering.

Macro Cluster	Txn Count	Gross Value (INR Cr)	Value Share (%)
Western Cluster	560,063	25.7	25.1%
Southern Cluster	569,311	25.7	25.1%
Eastern & North-East Cluster	564,192	25.6	25.0%
Northern Cluster	563,226	25.5	24.9%

Aggregating individual regions into macro-clusters provides strategic visibility into geographic revenue concentration for dark-kitchen network-investment and logistics planning.

7.0 Cross-Cutting Synthesis & Revenue Concentration Multipliers

Evaluating relationships across region and transaction type together reveals the specific product-market clusters driving disproportionate revenue.

Rank	Region	Transaction Type	Txn Count	Gross Value (INR Cr)
1	South Zone	Grocery / Instant Essentials	21,184	2.25
2	West Zone	Grocery / Instant Essentials	21,227	2.19
3	North Zone	Grocery / Instant Essentials	21,288	2.15
4	North-East Zone	Grocery / Instant Essentials	21,255	2.10
5	North-East Zone	Biryani & Rice Meals	47,180	2.05
6	West Zone	Biryani & Rice Meals	46,406	2.04
7	North-East Zone	North Indian Thali / Curries	42,446	2.00
8	North Zone	Biryani & Rice Meals	46,871	2.00
9	Metro Cluster B	Grocery / Instant Essentials	18,946	1.94
10	North Zone	North Indian Thali / Curries	42,581	1.94

Revenue Concentration: The High-Ticket Type Multiplier

The single strongest driver of top-line value is the concentration in high-value transaction types. Grocery / Instant Essentials and Biryani & Rice Meals together account for only 22.5% of transaction count but 30.2% of gross transaction value — a Pareto effect worth flagging for kitchen-capacity and pricing-strategy prioritization.

Geographic and transaction-type dimensions represent the most stable analytical axes in the dataset, while weekly trends reflect expected operational seasonality and end-of-period collection boundaries.

8.0 Payment Rail Preferences

Evaluating payment rails provides visibility into infrastructure dependency, interchange economics, and digital-payment adoption.

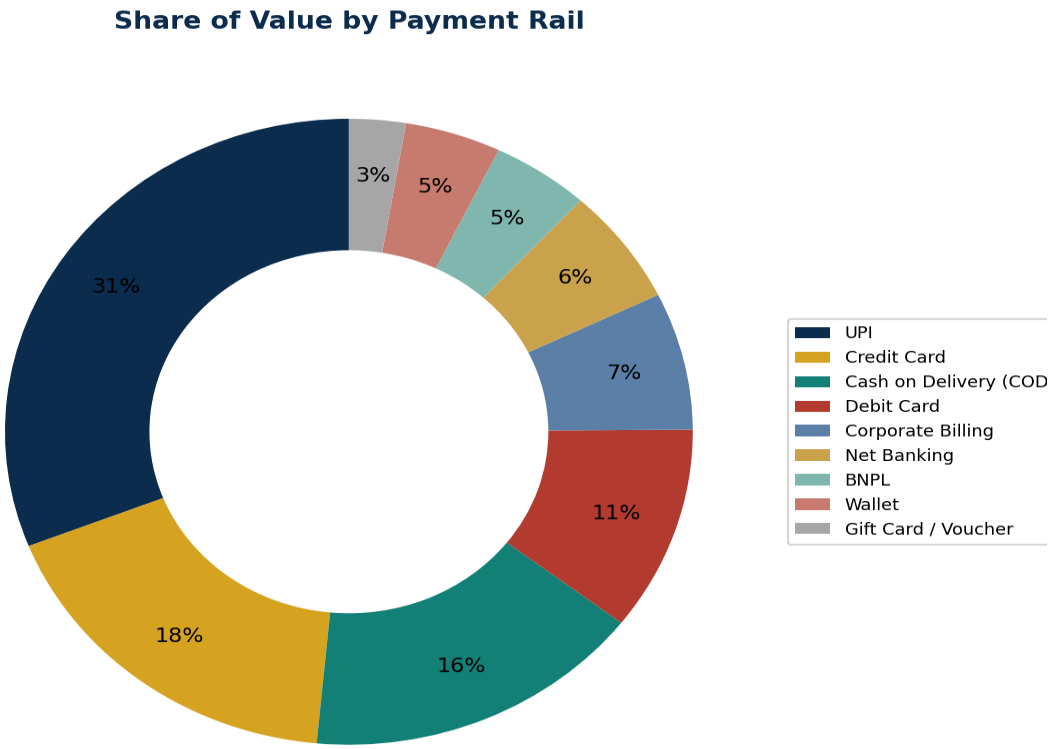


Figure 9. Share of Value by Payment Rail.

UPI dominates settlement value at 30.9% share, relative to its 33.2% share of transaction count.

Payment Rail	Txn Count	Value Share (%)	Count Share (%)
UPI	749,256	30.9%	33.2%
Credit Card	285,759	17.6%	12.7%
Cash on Delivery (COD)	436,556	16.0%	19.3%
Debit Card	240,006	10.6%	10.6%
Corporate Billing	79,901	7.1%	3.5%
Net Banking	125,677	6.0%	5.6%
BNPL	77,767	4.6%	3.4%
Wallet	174,326	4.5%	7.7%

Gift Card / Voucher	87,544	2.7%	3.9%

9.0 Transaction Success Rate & Reliability Analysis

Average success rate across all hub-days: **96.8%**, clustered tightly. In a real system this view is most useful segmented, not aggregated — hubs trending downward in rolling 7-day success rate are the earliest signal of a payment-gateway, restaurant-partner, or dispatch-system issue.

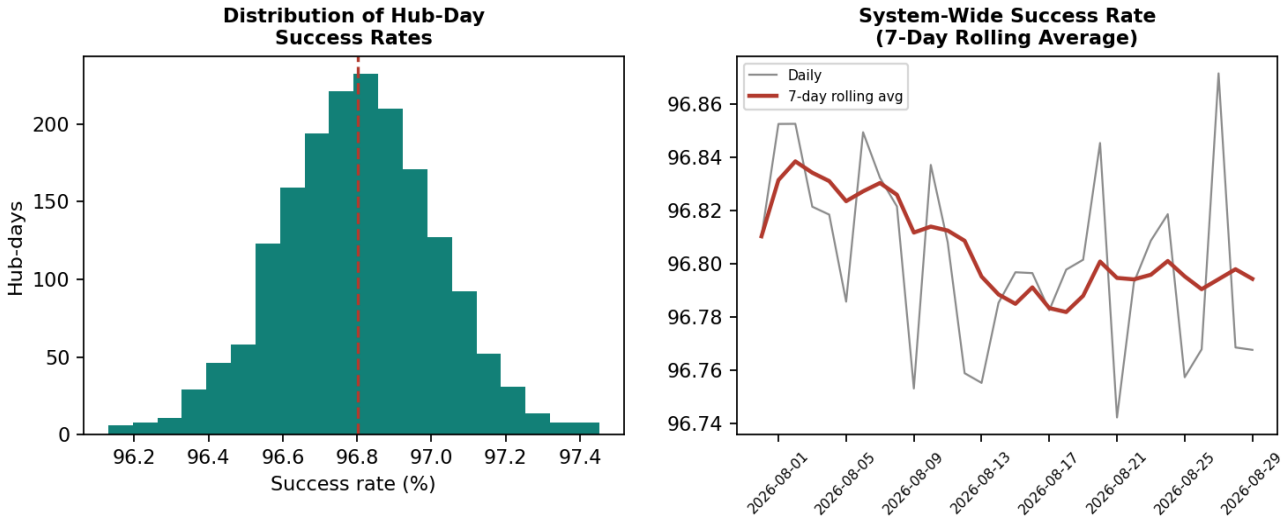


Figure 10. Hub-Day Success Rate Distribution and 7-Day Rolling Average.

Deeper Insight — Reliability Drift
The 7-day rolling average shown here has drifted downward over the observation window (96.81% → 96.79%) — worth a root-cause check before it compounds. Channels with high transaction volume but sub-97% success are the highest-risk combination, since volume amplifies the customer impact of every failed or cancelled order.

10.0 Rider Fleet Capacity & Delivery-Delay Risk

Hub-level rider fleet capacity was bucketed into three tiers:

- **Adequate Reserve** — comfortable buffer of available riders for expected order demand
- **Low Reserve (Reorder Zone)** — below 35% of capacity, rider-onboarding/shift trigger
- **Critical — Delivery Risk** — below 15% of capacity, cross-referenced against top-hub volume

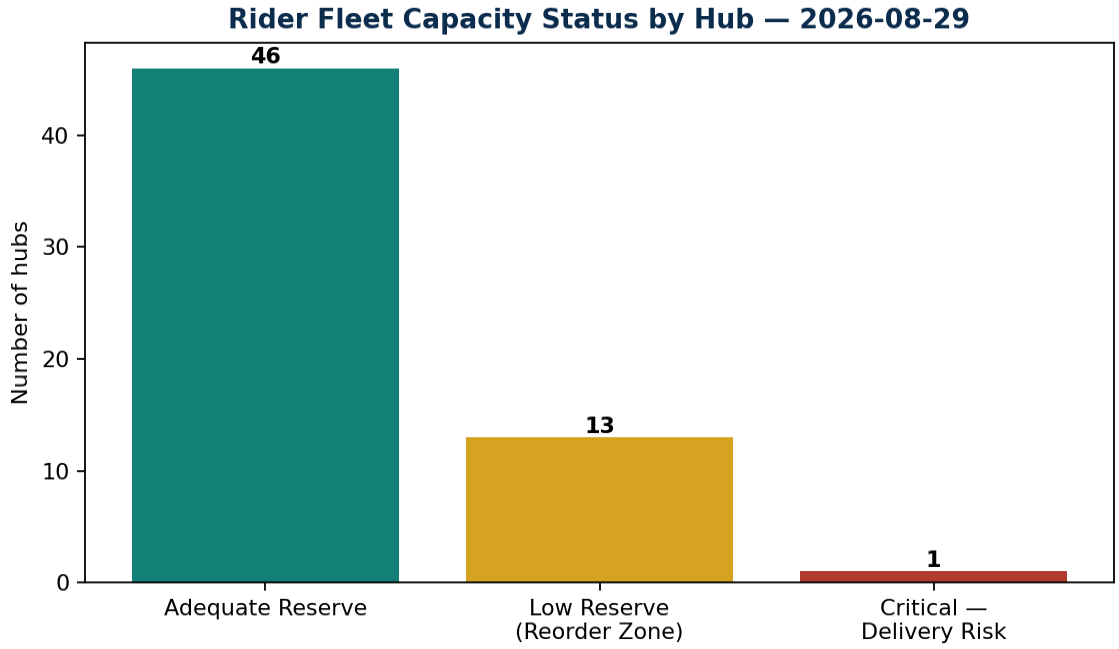


Figure 11. Rider Fleet Capacity Status by Hub — snapshot date.

Fleet-Logistics Recommendation
1 of 60 hubs were flagged critical and 13 more sat in the reorder zone on the most recent day. Production dashboards should cross-tab rider capacity against order-demand velocity to produce an “estimated delivery delay” metric per hub.

11.0 Volume, Value & Risk Elasticity Mechanics

Quantifying the mathematical relationships between transaction count, value, and reliability clarifies the key drivers of operational risk and top-line throughput.

Bivariate Variable Pair	Pearson Correlation (r)	Strength	Commercial Meaning
Txn Count vs. Gross Value	+0.79	Strong Positive	Hub-day value is substantially explained by transaction count, alongside type/seg
Txn Count vs. Success Rate	-0.04	Negligible Negative	Transaction volume shows little linear relationship with success rate at this aggr
Avg Value vs. Success Rate	+0.06	Negligible Positive	Average ticket size has little bearing on transaction reliability

Deeper Insight — Volume and Mix as Value Drivers

The positive correlation between transaction count and value (r = +0.79) shows that transaction count is an important lever for gross value, though transaction-type and segment mix add meaningful additional variation — consistent with a network where most hubs handle a broadly similar transaction mix but individual high-ticket types (bulk/party orders, grocery) still swing daily value.

12.0 Data Limitations, Scope & Assumptions

• **Dataset Scale:** The dataset contains ~2.26M simulated transactions across 60 delivery hubs. While ideal for descriptive-analytics demonstration, it represents an illustrative sample rather than a real platform's total national volume.

• **Observation Period:** The dataset runs across 30 days with a partial final week, which should not be interpreted as an ongoing demand contraction.

• **Absence of Customer-Level Data:** The transaction records are hub/region/type aggregates and do not include individual customer identifiers, restaurant-partner ratings, or dispute/refund records.

• **Descriptive Non-Causal Scope:** Findings such as Grocery / Instant Essentials's value leadership represent observed patterns within this dataset, not proof that transaction type alone drives fleet-capacity risk.

• **Nominal Currency Reporting:** All financial figures reflect nominal INR as recorded, without adjustment for inflation or promotional discounting.

Data Validation & Governance

All aggregated rows were verified for schema consistency, non-negative values, and mathematical consistency (count x avg value = total value). Zero anomalies or negative values were found.

13.0 Next Steps: Move From Descriptive → Predictive Analytics

Descriptive analytics explains past transaction outcomes. The next analytical step is leveraging these empirical patterns to train predictive models that forecast future demand and risk.

Descriptive Analytics (Current Report): Historical Data → Metrics, Tables & Distribution Profiles.
Predictive Analytics (Next Phase): Historical Baselines → Machine Learning Models & Demand/Risk Forecasts.
Prescriptive Analytics (Advanced Phase): Predictions → Automated Rider Dispatch, Dynamic Fraud Rules & Delivery Routing.

Predictive Target & Objective	Candidate ML Models	Key Input Features
Demand forecasting per hub	ARIMA, SARIMAX, Prophet, LSTM	Historical order volume, calendar effects, weather, promo dates
Real-time fraud / anomaly detection	Isolation Forests, Autoencoders, XGBoost	Transaction velocity, amount deviation, channel, time-of-day
Success-rate degradation early warning	Change-point detection, Random Forest	Rolling success rate, hub load, gateway latency
Customer cohort & retention modeling	Gradient Boosted Regressors, survival models	Transaction recency, frequency, segment, channel mix

Happy to build out any of these next, or swap in a real dataset (order-management export, a data warehouse connection, etc.) if you have one — the same pipeline and chart structure above would apply directly.

14.0 Summary of Key Metrics & Executive Scorecard

Metric Area	Top Segment / Value	Metric Value	Commercial Takeaway
Gross Transaction Value	Entire Dataset	INR 102 Cr	Total value across 30 days
Total Transaction Volume	All Transaction Types	2.26M txns	Aggregate volume across 10 types
Top Type by Value	Grocery / Instant Essentials	15.9% share	Leading value contributor
Top Type by Volume	Fast Food & Burgers	16.0% share	Leading count contributor
Top Geographic Region	North-East Zone	13.4% share	Largest single-region market
Top Customer Segment	Regular Customers	41.8% share	Largest value-contributing segment
Dominant Channel	Mobile App	66.9% share	Primary transaction-initiation channel
Average Success Rate	All Hubs	96.8%	System-wide transaction reliability
Hubs at Fleet Capacity Risk	Low + Critical Reserve	14 of 60	Fleet-logistics attention needed

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Dataset Citation: Online Food Delivery Daily Transaction Dataset (60 hubs x 10 types x 8 regions x 30 days, ~2.26M simulated transactions): synthetic project dataset recording Date, Hub_ID, Region, Transaction_Type, Transaction_Count, Total_Value, Success_Rate, Declined_Count, and Rider_Capacity_Pct.

Technical Appendix — Schema & Formulas

Appendix A: Database Schema & Field Data Dictionary

Field Name	Data Type	Sample Value	Description
date	Date	2026-07-31	Transaction date (YYYY-MM-DD)
hub_id	Categorical	HUB033	Delivery hub identifier (60 unique hubs)
region	Categorical	South Zone	Geographic region (8 unique zones)
txn_type	Categorical	Biryani & Rice Meals	Transaction type (10 distinct types)
txn_count	Integer	1,204	Number of transactions in this row
total_value_inr	Float	1,024,500.00	Aggregate value in INR
success_rate	Float	0.968	Share of transactions completed successfully
declined_count	Integer	12	Number of declined/failed transactions
rider_capacity_pct	Float	0.55	Hub rider fleet capacity as a fraction of full strength

Appendix B: Aggregation Formulations

- Gross Transaction Value: Sum(total_value_inr) across all rows
- Average Value per Transaction: Gross Value / Total Transaction Count
- Type / Segment / Region Value Share (%): (Category Value / Total Gross Value) x 100
- Success Rate (weighted): Weighted average of success_rate, weighted by txn_count

Glossary of Food-Delivery Analytics Terms

Term	Definition & Commercial Context
Descriptive Analytics	Summarizing and visualizing historical transaction data to understand past operational performance without predictive modeling
Average Transaction Value (ATV)	A food-delivery operations KPI measuring the average value moved each time a transaction completes.
Revenue/Value Concentration	An distribution pattern where a small number of transaction types or regions generate a disproportionate share of total value.
Interquartile Range (IQR)	The statistical spread between the 25th (Q1) and 75th (Q3) percentiles, capturing the middle 50% of values.
Success Rate	The share of initiated transactions that complete successfully, net of declines, cancellations and technical failures.
Rider Fleet Capacity / Delivery Delay Risk	The risk a delivery hub's active rider fleet runs below the buffer needed to meet expected order demand.
Pearson Correlation (r)	A statistical coefficient measuring the linear relationship between two numerical variables.
Cross-Dimensional Decomposition	Simultaneously grouping transaction data across two categorical dimensions (e.g. Region x Transaction Type) to evaluate granular perf
Predictive Transition	Using historical transaction patterns to engineer features and train supervised ML models for demand or risk forecasting.

Report Conclusion

This concludes the Online Food Delivery Transaction Descriptive Analytics Report. All analyses, tables, charts, and technical appendices are fully documented and validated for review.